

The Midgap Circle for Normal Packet Blocks

An Exact Gap–Feedback Criterion and Its Nonnormal Boundary

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Abstract

The Schur theorem of RH-163 reduces packet-to-Riesz rank preservation to $adb c < 1$, but leaves the contour and two resolvent bounds unspecified. We solve this geometric problem for a benchmark class. Suppose the packet and complement blocks are normal, the packet spectrum lies in $|z - \mu| \leq \rho$, and the complementary spectrum lies in $|z - \mu| \geq R > \rho$. Among centered circles, the radius

$$s_* = (\rho + R)/2$$

minimizes the Schur feedback. The optimized rank condition is exactly

$$2\sqrt{bc} < R - \rho.$$

We also give the resulting projector envelope and prove sharpness of the midpoint for the available disk data. A 512-system finite audit has zero rank-count and resolvent-bound failures. The theorem deliberately stops at normal blocks: for a nonnormal operator, spectral distance alone cannot replace a resolvent estimate. No physical normality or all-level pseudospectral theorem is claimed.

1 Disk-separated normal blocks

Let

$$A = \begin{pmatrix} A_P & B \\ C & A_Q \end{pmatrix}$$

on an orthogonal packet decomposition. Assume A_P and A_Q are normal and, for a center $\mu \in \mathbb{C}$,

$$\sigma(A_P) \subset \{|z - \mu| \leq \rho\}, \quad \sigma(A_Q) \subset \{|z - \mu| \geq R\}, \quad R > \rho.$$

Put $b = \|B\|$ and $c = \|C\|$. For $\rho < s < R$, let $\Gamma_s = \{|z - \mu| = s\}$.

Normality gives the exact spectral-distance resolvent formula. Therefore

$$a(s) := \sup_{\Gamma_s} \|(z - A_P)^{-1}\| \leq \frac{1}{s - \rho}, \quad d(s) := \sup_{\Gamma_s} \|(z - A_Q)^{-1}\| \leq \frac{1}{R - s}.$$

2 Optimal midpoint theorem

Theorem 1 (Midgap feedback optimum). *Among centered circles Γ_s , the upper bound*

$$\kappa(s) = \frac{bc}{(s - \rho)(R - s)}$$

is minimized uniquely at

$$s_* = (\rho + R)/2,$$

where

$$\kappa_* = \frac{4bc}{(R - \rho)^2}.$$

Consequently Γ_{s_} certifies packet-rank preservation whenever*

$$\boxed{2\sqrt{bc} < R - \rho.}$$

Proof. The denominator is

$$(s - \rho)(R - s) = \frac{(R - \rho)^2}{4} - \left(s - \frac{R + \rho}{2}\right)^2.$$

It is uniquely maximized at the midpoint. Substitution gives κ_* . If $\kappa_* < 1$, the RH-163 Schur homotopy theorem preserves the enclosed rank [Program, 2026]. $\square$

The constant is sharp for this disk information. Scalar packet and complement blocks placed at the two nearest radial boundary points attain both resolvent denominators. At $4bc = (R - \rho)^2$, a two-mode matrix can have an eigenvalue on the midpoint circle.

3 Projector envelope

Let $g = R - \rho$, $s_* = (R + \rho)/2$, and $\kappa = 4bc/g^2 < 1$. At the midpoint, $a = d = 2/g$. The scalar block estimate of RH-163 yields

$$\|\Pi - P\| \leq \frac{s_*}{1 - \kappa} \left\| \begin{pmatrix} 2\kappa/g & 4b/g^2 \\ 4c/g^2 & 2\kappa/g \end{pmatrix} \right\|_2.$$

This graph bound depends on the absolute contour radius s_* as well as the gap. Moving both spectral sets far from the origin leaves the rank gate unchanged but lengthens the circle; a noncircular contour or a directional graph estimate may then be preferable.

4 Why the theorem stops at normality

For a normal operator, $\|(z - A)^{-1}\| = \text{dist}(z, \sigma(A))^{-1}$. For a nonnormal matrix, the resolvent can be arbitrarily larger while the spectrum and radial gap are fixed [Trefethen and Embree, 2005]. A Jordan block already exhibits polynomial resolvent amplification. Therefore the

condition $2\sqrt{bc} < g$ is not a physical prime-dynamics theorem unless normality, similarity conditioning, or an independent pseudospectral envelope is proved.

One can retain the geometry without normality by replacing ρ and R with certified pseudospectral barriers that directly imply the displayed resolvent bounds. That replacement is an input, not a consequence of the eigenvalue plot.

5 Conditioned pseudospectral inflation

There is a useful intermediate statement between normality and a completely free resolvent. Suppose that on every centered circle one has certified envelopes

$$\sup_{\Gamma_s} \|(z - A_P)^{-1}\| \leq \frac{K_P}{s - \rho}, \quad \sup_{\Gamma_s} \|(z - A_Q)^{-1}\| \leq \frac{K_Q}{R - s},$$

where $K_P, K_Q \geq 1$ are fixed conditioning factors.

Corollary 2 (Inflated midgap gate). *The midpoint circle remains optimal for these envelopes and certifies rank whenever*

$$2\sqrt{K_P K_Q bc} < R - \rho.$$

Proof. The Schur product is bounded by $K_P K_Q bc / [(s - \rho)(R - s)]$. The same quadratic denominator as in Theorem 1 is maximized at the midpoint, where the product is $4K_P K_Q bc / (R - \rho)^2$. $\square$

This corollary quantifies the price of nonnormality when a uniform similarity or pseudospectral constant is available. If $K_P K_Q$ grows with scale, the geometric gap may widen while the usable margin still collapses; both factors must therefore be recorded in the physical ledger.

6 Audit and route consequence

The audit samples 512 diagonal normal block pairs with complex spectra in the two radial regions. Directed couplings are scaled so the midpoint feedback is below 0.55. On 1024 contour nodes it verifies both resolvent bounds, and direct diagonalization confirms exactly the packet rank inside the circle. No failure occurs. Floating-point rank counts are illustrative regression checks, not interval proofs.

RH-165 removes contour optimization from the normal benchmark and supplies a sharp geometric falsifier. The physical family remains nonnormal, so the next steps must obtain directed Ritz residuals and then certify continuous resolvent envelopes. The paper does not close physical R, Gate A, or any later Hilbert–Polya interface.

References

Prime Dynamics Theory Program. A two-sided schur certificate for packet-to-riesz lifting, 2026. RH-163 manuscript.

Lloyd N. Trefethen and Mark Embree. *Spectra and Pseudospectra*. Princeton University Press, 2005.